p-book: Investigating Navigation Preferences in Personalized Educational Books

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ABSTRACT

We present p-book, a web-based system that transforms linear educational content into a personalized, multi-modal reading experience. The system implements a *spine+depth* content architecture where a mandatory narrative backbone is augmented with optional depth blocks personalized to reader engagement preferences. p-book offers four navigation paradigms—Missions, Browse, Read, and Map—over the same content graph. We instrument all interactions to study: *which navigation paradigm do readers prefer, and how do preferences relate to engagement and coverage?* We describe the architecture, its grounding in adaptive hypermedia and self-determination theory, and our planned study.

CCS CONCEPTS

• **Human-centered computing** → **Interactive systems and tools**; • **Information systems** → **Recommender systems**.

KEYWORDS

personalized book, adaptive hypermedia, navigation preferences, recommender systems, user modeling

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1 INTRODUCTION

Adaptive learning systems have demonstrated substantial effectiveness (g=0.70 on cognitive outcomes [22]), yet the vast majority—Carnegie Learning [13], ALEKS [6], Duolingo [18]—focus on skill-based tutoring and drill exercises. Book-length narrative educational content has received surprisingly little attention from the adaptive hypermedia community [20].

We present **p-book**, a system that brings adaptive hypermedia to narrative reading through a *spine+depth* content model: a linear backbone of core content (spine) is augmented with optional depth blocks that readers choose based on engagement preferences. A distinctive feature is that p-book offers **four navigation paradigms**—Missions, Browse, Read, and Map—all over the same content graph, following self-determination theory: meaningful choices support autonomy and intrinsic motivation [14], with 3–5 options being optimal [8, 16].

The **primary research question** is: *Which navigation paradigm do readers prefer, and how do preferences relate to engagement depth and content coverage?* We instrument all interactions to collect behavioral data addressing this question. Whether personalization improves learning outcomes is deferred to a planned controlled study.

Table 1: Architectural comparison of adaptive educational systems.

	Carnegie	*ALEKS*	*Duolingo*	*Knewton*	*LECTOR*	***p-book***
Content	skill	skill	drill	skill	linear	**spine+d.**
User model	mastery	mastery	mastery	mastery	behav.	**engag.**
Adaptation	rules	KST	HLR	prob.	obs.	**RecSys**
Agency	none	none	none	low	none	**4 modes**
Spaced rep.	no	no	yes	no	no	**yes**
Content type	exerc.	exerc.	drills	exerc.	text	**narrat.**

2 RELATED WORK

2.1 Adaptive Platforms and Intelligent Tutoring

Table 1 compares p-book against major adaptive learning systems. Carnegie Learning uses ACT-R cognitive theory and Bayesian knowledge tracing to select practice problems [7, 13]. ALEKS employs Knowledge Space Theory for adaptive math exercises [6]. Knewton Alta adapts textbook exercises using probabilistic models [5]. These systems share a common architecture: content decomposed into skill atoms, mastery-based user models, and system-controlled paths.

Duolingo [18] is the closest multi-component precedent, combining adaptive sequencing, spaced repetition via half-life regression, and gamification. However, it operates on vocabulary drills in a system-controlled sequence—it supports neither reader-directed navigation nor narrative content. The p-book is unique in three respects: (1) adaptation for narrative/conceptual text; (2) six recommendation scenarios mapped to pedagogical functions; and (3) four reader-controlled navigation modes.

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2.2 Adaptive Hypermedia and Intelligent Textbooks

The adaptive hypermedia tradition [2] established adaptive navigation support, presentation, and curriculum sequencing. Sosnovsky and Brusilovsky [19] demonstrated topic-based adaptation in the UMUAI journal. The intelligent textbook initiative [20] explores concept extraction and adaptive navigation but without integrating recommendation engines, spaced repetition, or gamification. LECTOR [10] tracks e-book reading behavior to build learner models but is observational—it does not recommend content or adapt the path.

2.3 Engagement Modes, Not Learning Styles

p-book offers three engagement modes—Explorer (*"Show me!"*), Creator (*"Let me build!"*), Thinker (*"Why?"*)—influencing depth-block recommendations. We explicitly distinguish these from learning styles, whose matching to instruction is decisively debunked [11, 21]. Unlike the "meshing hypothesis," engagement modes are *dynamic preference priors updated from behavioral signals*: a reader who selects Explorer but engages with Thinker content sees recommendations adapt. This is grounded in SDT [14] (choice supports autonomy), interest-driven learning [17] (choice increases engagement through perceived autonomy), and elaboration theory [12] (Explorer/Creator/Thinker map to relational/procedural/conceptual elaboration types).

2.4 Gamification and Spaced Repetition

Gamification yields moderate cognitive effects (g=0.49) [15] but shows novelty decline after ∼4 weeks [9], and negative effects when creating anxiety [1]. p-book's gamification is deliberately conservative: XP and badges, no leaderboards or streaks. Spaced repetition substantially improves retention [4], with synergistic effects when combined with gamification in K-12 reading [23].

3 SYSTEM ARCHITECTURE

3.1 Spine+Depth Content Model

Content is structured as a directed graph with two node types. **Spine blocks** form the mandatory narrative backbone. **Depth blocks** branch off spine nodes, tagged by engagement mode and content type (sidebar, game, worksheet, extended explanation). The current instantiation, *"How Recommendations Work"* for ages 8–15, contains 83 blocks across 6 chapters: 28 spine, 20 depth, 22 sidebars, 10 games, and 3 worksheets. The spine guarantees coverage structurally—no assessment gating required.

3.2 Four Navigation Paradigms

The same content graph is navigable through four modes:

- **Missions:** Story-driven quests sequencing 5–8 blocks into narrative arcs with culminating challenges. Most structured.
- **Browse:** Netflix-style carousels organized by recommendation scenario ("For You," "Popular," "Continue Reading"). Most discovery-oriented.
- **Read:** Linear chapter progression with expandable depth cards. Most familiar.
- **Map:** Visual overview of all chapters and progress. Most exploratory.

Readers choose an initial mode during onboarding and can switch freely; switches are logged as first-class events.

3.3 Recommendation and Instrumentation

Personalization is powered by Recombee through six scenarios mapping to pedagogical functions: `next-in-spine` (curriculum sequencing), `go-deeper` (adaptive elaboration), `related` (lateral connections), `for-you` (personalized discovery), `popular` (social proof), and `start-here` (cold-start onboarding). The model receives detail views with duration, ratings, bookmarks, and engagement-mode affinity scores. A local fallback operates when the API is unavailable.

All interactions are instrumented: navigation events and mode switches, dwell time and scroll depth per block, depth-block expansions by engagement mode, recommendation acceptance with scenario attribution, and spaced-repetition responses. This enables behavioral analysis without external instruments.

4 STUDY DESIGN

We aim to answer four research questions through log analysis: **RQ1:** Which navigation paradigm do readers prefer, and are preferences stable? **RQ2:** How do paradigms differ in coverage and engagement depth? **RQ3:** Do multi-mode users achieve broader coverage? **RQ4:** Does recommendation increase depth-block exploration (within-subject, recommendations on vs. off)?

We plan a within-subjects deployment where participants use p-book freely over multiple sessions. For RQ4, we toggle recommendation visibility. All dependent variables—mode dwell time, switch frequency, spine completion, depth exploration rate, recommendation CTR—are computed from logs.

During internal piloting (N=8, adults), all four modes were used; Browse and Read were most popular initially; Map attracted users who explored more depth blocks; mission completion was high once started but discovery was low without prompting. These preliminary observations will be validated with the target age group.

5 CONCLUSIONS

p-book applies adaptive hypermedia to narrative educational books via a spine+depth architecture, multi-scenario recommendation, and four navigation paradigms. The primary contribution is the instrumented investigation of *navigation preferences* in personalized educational content—an understudied question in adaptive learning. Our immediate goal is a within-subjects study of navigation preference and engagement. Longer-term, we plan controlled evaluation of personalization effects on comprehension and retention with children aged 8–15.

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